

IT Team — Server & Infrastructure Setup Guide

Project: Self-hosted Email Sending Platform

Prepared: 2026-06-24

Target volume: ~1,000,000 emails/day

Estimated setup time: 2-3 days (excluding IP warmup period)

Table of Contents

- [1. Overview — What We Need](#)
 - [2. Option A: OVH Setup](#)
 - [3. Option B: Hetzner Setup](#)
 - [4. Domain & DNS Configuration](#)
 - [5. IP Address Setup](#)
 - [6. Port 25 — Unblock SMTP](#)
 - [7. SSL Certificates](#)
 - [8. Feedback Loop Registration](#)
 - [9. Server Software Requirements](#)
 - [10. Handoff Checklist](#)
 - [11. Cost Estimates](#)
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1. Overview — What We Need

We are deploying a self-hosted email sending platform. The IT team needs to procure and configure:

Item	Quantity	Purpose
Dedicated server	1	Runs all application containers (not a VPS — we need unthrottled sending)
IP addresses	5-10 to start (up to 30 later)	Sending IPs for email delivery, separated by stream
Domain	1 root domain (with multiple subdomains)	Sending identity (SPF, DKIM, DMARC)
Port 25 outbound	Must be unblocked	Required for direct-to-MX email delivery
SSL certificate	1 (Let's Encrypt wildcard or per-subdomain)	HTTPS for web UI + tracking

Server specifications (minimum)

Spec	Minimum	Recommended
CPU	4 cores	8 cores
RAM	16 GB	32 GB
Storage	200 GB SSD	500 GB NVMe
Bandwidth	1 Gbps unmetered	1 Gbps unmetered
OS	Ubuntu 22.04 / 24.04 LTS	Ubuntu 24.04 LTS

Critical: Must be a **dedicated server** (bare metal), NOT a budget VPS. Budget VPS providers throttle outbound SMTP and share IPs.

2. Option A: OVH Setup

2.1 Create OVH Account

1. Go to <https://www.ovhcloud.com>
2. Create a business account (not personal — needed for IP blocks)
3. Verify identity (may require business documents)
4. Add payment method

2.2 Order Dedicated Server

1. Navigate to **Bare Metal Cloud** → **Dedicated Servers**
2. Select a server matching our specs:
 - Recommended: **Advance** tier or higher
 - Location: Choose based on target audience geography
 - **Avoid Rise/Essential tiers** — limited bandwidth
3. Select **Ubuntu 24.04** as the OS
4. Complete order

Note down:

- Server IP (main IP)
- Server hostname
- Root SSH credentials

2.3 Order IP Block

1. In OVH Manager → **Bare Metal Cloud** → your server → **IP**
2. Click "**Order additional IPs**"

3. Order a **/29 block** (5 usable IPs) to start
 - Or individual IPs if blocks aren't available
4. Later scale to **/28** (13 usable IPs) or **/27** (29 usable IPs)

Important: Select IPs from the **same datacenter** as your server.

2.4 Set PTR Records (Reverse DNS)

For **each** IP address:

1. OVH Manager → **Bare Metal Cloud** → **IP** → find the IP
2. Click the **gear icon** → "**Modify reverse DNS**"
3. Set the reverse DNS to match the A record hostname:

IP	PTR Record (Reverse DNS)
1st IP	mail.yourdomain.com
2nd IP	mail.yourdomain.com (or mail2.yourdomain.com)
3rd IP	eng.yourdomain.com
4th IP	out.yourdomain.com
5th IP	out2.yourdomain.com

Rule: The PTR record **MUST** match the A record. Set A records first (Section 4), wait for propagation, then set PTR.

2.5 OVH API (optional — for automation)

OVH provides an API for managing IPs and PTR records:

- API console: <https://api.ovh.com/console/>
- Useful for bulk PTR setup across many IPs
- Credentials: Application Key + Application Secret + Consumer Key

3. Option B: Hetzner Setup

3.1 Create Hetzner Account

1. Go to <https://www.hetzner.com>
2. Create account → verify identity
3. Add payment method

3.2 Order Dedicated Server

1. Navigate to **Dedicated** → **Server Auction** (best value) or **Configurator**

2. Recommended specs:
 - **AX42** or similar (AMD Ryzen, 64GB RAM, NVMe)
 - Location: Falkenstein or Helsinki
3. Select **Ubuntu 24.04**
4. Complete order

3.3 Order IP Block

1. Hetzner Robot → **Server** → **IPs** → **Order additional IPs/subnets**
2. Request a **/29 subnet** (5 usable IPs)
3. Hetzner may ask for justification — respond:

"We are setting up a self-hosted email sending platform. We need separate IPs for different mail streams (transactional, marketing, cold outreach) to maintain reputation isolation."

3.4 Set PTR Records

1. Hetzner Robot → **Server** → **IPs**
2. Click each IP → "**Edit reverse DNS**"
3. Set PTR to match A records (same table as OVH section above)

Note: Hetzner's clean IP reputation is a major advantage. Their IPs are generally less abused than OVH's.

4. Domain & DNS Configuration

4.1 Subdomains to Create

We use **separate subdomains per sending stream** for reputation isolation:

Subdomain	Stream	Purpose
mail.yourdomain.com	Opt-in	Legitimate subscriber emails
eng.yourdomain.com	Engaged	Openers/clickers from cold list
out.yourdomain.com	Cold	Never-engaged, cold outreach
postal.yourdomain.com	—	Postal web UI
track.yourdomain.com	—	Open/click tracking

4.2 A Records

Create these A records at your domain registrar (GoDaddy, Cloudflare, etc.):

Type	Hostname	Value	TTL
A	mail.yourdomain.com	IP_ADDRESS_1	3600
A	eng.yourdomain.com	IP_ADDRESS_3	3600
A	out.yourdomain.com	IP_ADDRESS_4	3600
A	postal.yourdomain.com	SERVER_MAIN_IP	3600
A	track.yourdomain.com	SERVER_MAIN_IP	3600

Create A records FIRST, wait 1-2 hours for propagation, THEN set PTR records.

4.3 SPF Records

One SPF record per sending subdomain:

Type	Hostname	Value
TXT	mail.yourdomain.com	v=spf1 ip4:IP_ADDRESS_1 ip4:IP_ADDRESS_2 ~all
TXT	eng.yourdomain.com	v=spf1 ip4:IP_ADDRESS_3 ~all
TXT	out.yourdomain.com	v=spf1 ip4:IP_ADDRESS_4 ip4:IP_ADDRESS_5 ~all

Include all IPs that will send from each subdomain.

4.4 DKIM Records

DKIM public keys will be generated by the Postal mail server during setup. After Postal is deployed:

1. Log into Postal UI → each mail server → Domain → DKIM
2. Copy the DKIM public key
3. Create TXT record:

Type	Hostname	Value
TXT	postal-optin._domainkey.mail.yourdomain.com	v=DKIM1; k=rsa; p=GENERATED_PUBLIC_KEY
TXT	postal-engaged._domainkey.eng.yourdomain.com	v=DKIM1; k=rsa; p=GENERATED_PUBLIC_KEY

Type	Hostname	Value
TXT	postal-cold._domainkey.out.yourdomain.com	v=DKIM1; k=rsa; p=GENERATED_PUBLIC_KEY

4.5 DMARC Record

One DMARC record at the **org domain level**:

Type	Hostname	Value
TXT	_dmarc.yourdomain.com	v=DMARC1; p=quarantine; rua=mailto:dmarc@yourdomain.com; pct=100

Start with **p=quarantine**. Move to **p=reject** after monitoring shows clean results.

4.6 MX Records (for receiving bounces)

Type	Hostname	Value	Priority
MX	mail.yourdomain.com	mail.yourdomain.com	10
MX	eng.yourdomain.com	eng.yourdomain.com	10
MX	out.yourdomain.com	out.yourdomain.com	10

4.7 DNS Propagation

- A records: 1-4 hours
- TXT records (SPF/DKIM/DMARC): 1-4 hours
- PTR records: Usually instant at OVH/Hetzner, up to 24 hours elsewhere

Verify with:

```
# Check A record
dig mail.yourdomain.com A +short

# Check SPF
dig mail.yourdomain.com TXT +short

# Check DMARC
dig _dmarc.yourdomain.com TXT +short

# Check PTR
dig -x IP_ADDRESS +short

# Full email auth check
# Use https://mxtoolbox.com/SuperTool.aspx
```

5. IP Address Setup

5.1 IP Allocation Plan

IP	Stream	Type	Subdomain	Daily Cap (warmup start)
IP #1	Opt-in	Dedicated	mail.yourdomain.com	500
IP #2	Opt-in	Dedicated	mail.yourdomain.com	500
IP #3	Engaged	Dedicated	eng.yourdomain.com	200
IP #4	Cold	Semi-disposable	out.yourdomain.com	100
IP #5	Cold	Semi-disposable	out.yourdomain.com	100

5.2 Shared vs Dedicated

Type	Meaning	When to use
Dedicated	One IP per domain/stream, protected	Opt-in stream (most valuable)
Shared	Multiple domains on one IP	Only if running multiple brands
Semi-disposable	Expect occasional blocklisting	Cold stream — isolated from opt-in

5.3 IP Configuration Checklist (per IP)

- ☐ IP purchased and assigned to server
- ☐ A record created pointing subdomain → IP
- ☐ PTR record set matching A record hostname
- ☐ IP registered in CRM → IP Pools
- ☐ IP assigned to correct stream and pool
- ☐ Daily warmup cap configured

6. Port 25 — Unblock SMTP

By default, most providers block outbound port 25 to prevent spam from compromised servers.

OVH

1. Log into OVH Manager
2. Go to **Bare Metal Cloud** → your server
3. Check if port 25 is open: **Network** → **Firewall**
4. If blocked, open a **support ticket**:
 - Subject: "Request to unblock outbound port 25 for email sending"
 - Body:

We are setting up a legitimate email marketing platform on server [SERVER_NAME]. We need outbound port 25 unblocked for direct-to-MX email delivery. We will implement proper authentication (SPF, DKIM, DMARC), feedback loops, and unsubscribe handling. Our sending domains are: mail.yourdomain.com, eng.yourdomain.com, out.yourdomain.com.

Typical response time: 1-3 business days. Do this early.

Hetzner

1. Hetzner **does not block port 25 on dedicated servers** by default
2. Verify: SSH into server and run:

```
telnet gmail-smtp-in.l.google.com 25
```

If it connects, port 25 is open.
3. If blocked (rare for dedicated), open a support ticket with same justification.

Verification

From the server, test connectivity:

```
# Test connection to Gmail
telnet gmail-smtp-in.l.google.com 25

# Test connection to Outlook
telnet outlook-com.olc.protection.outlook.com 25

# Expected: "220" greeting from the remote server
# If timeout: port 25 is blocked
```

7. SSL Certificates

What needs SSL

Service	Domain	Why
Postal Web UI	postal.yourdomain.com	Admin interface HTTPS
Tracking	track.yourdomain.com	Open/click tracking links
CRM	crm.yourdomain.com	Your CRM frontend

Service	Domain	Why
	(optional)	

Using Let's Encrypt (free)

```
# Install certbot on the server
sudo apt install certbot

# Generate certificates
sudo certbot certonly --standalone -d postal.yourdomain.com -d
track.yourdomain.com

# Certificates will be at:
# /etc/letsencrypt/live/postal.yourdomain.com/fullchain.pem
# /etc/letsencrypt/live/postal.yourdomain.com/privkey.pem

# Auto-renewal is configured automatically
```

8. Feedback Loop Registration

Register for feedback loops so spam complaints flow back to our system for automatic suppression.

Microsoft SNDS (Smart Network Data Services)

1. Go to <https://sendersupport.olc.protection.outlook.com/snds/>
2. Sign in with a Microsoft account
3. Add your sending IPs
4. Verify ownership (they check PTR records)
5. This gives you visibility into how Microsoft views your IPs

Microsoft JMRP (Junk Mail Reporting Program)

1. Go to <https://sendersupport.olc.protection.outlook.com/snds/JMRP.aspx>
2. Register your IPs
3. Complaints from Outlook/Hotmail users will be sent to your `abuse@yourdomain.com`

Google Postmaster Tools

1. Go to <https://postmaster.google.com>
2. Add and verify your sending domain
3. Monitor: spam rate, authentication, IP reputation, delivery errors

Required email addresses

Create these mailboxes (or forwarding addresses):

Address	Purpose
abuse@yourdomain.com	Receives spam complaints from feedback loops
postmaster@yourdomain.com	Required by RFC, receives delivery issues
dmarc@yourdomain.com	Receives DMARC aggregate reports

9. Server Software Requirements

The development team will handle deployment, but the server needs these pre-installed:

```
# Update system
sudo apt update && sudo apt upgrade -y

# Install Docker
curl -fsSL https://get.docker.com | sh
sudo usermod -aG docker $USER

# Install Docker Compose (included with modern Docker)
docker compose version

# Install useful tools
sudo apt install -y htop iotop net-tools dnsutils telnet
```

That's all the IT team needs to install. The application runs entirely in Docker containers — the dev team deploys using `docker compose up -d`.

10. Handoff Checklist

When the IT team is done, provide the dev team with:

Item	Details
Server SSH access	IP, username, SSH key or password
Server main IP	The primary IP of the server
IP block	List of all sending IPs (e.g., 192.168.1.10-14)
Domain	The root domain being used
DNS access	Credentials for the domain registrar (or delegate a zone)
Port 25 status	Confirmed open or ticket number if pending
SSL certs	Path to certificates on server
Provider login	OVH/Hetzner panel credentials for IP/PTR management
Feedback loop status	SNDS, JMRP, Postmaster Tools — registered or pending

Verification commands the dev team will run

```
# Check Docker
docker --version
docker compose version

# Check port 25
telnet gmail-smtp-in.l.google.com 25

# Check DNS
dig mail.yourdomain.com A +short
dig mail.yourdomain.com TXT +short
dig _dmarc.yourdomain.com TXT +short
dig -x SENDING_IP +short

# Check disk/RAM
free -h
df -h
```

11. Cost Estimates

OVH (monthly)

Item	Cost (approx)
Dedicated server (Advance)	€60-90/month
IP block (/29 = 5 IPs)	€10-15/month
Additional IPs (each)	€2-3/month
Total (starting)	~€75-110/month

Hetzner (monthly)

Item	Cost (approx)
Dedicated server (AX42)	€49-65/month
IP block (/29 = 5 IPs)	€5-10/month
Additional IPs (each)	€1-2/month
Total (starting)	~€55-80/month

Hetzner is typically 20-30% cheaper with cleaner IPs. Either works — the software is identical.

Document prepared for IT team. Contact the development team for deployment after infrastructure is ready.